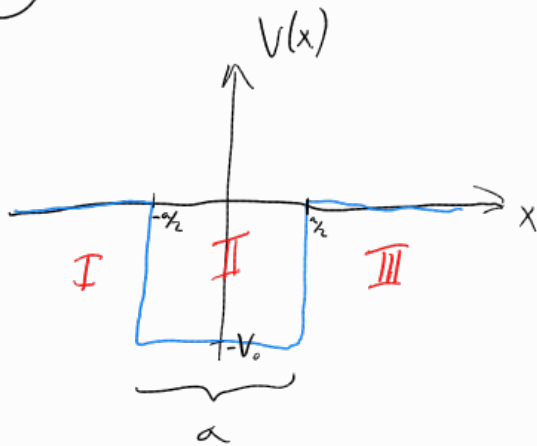


$$1D \longrightarrow x$$

$$\text{klasirno: } x(t) \quad x(e), \dot{x}(0) \rightarrow 2\pi\hbar \rightarrow x(t)$$

$$\text{kvantno: } \Psi(x, t) \quad \Psi(x, 0) \rightarrow SE \rightarrow \Psi(x, t)$$

①



$E < 0$ (vezana lastna stanja)

$$H = \frac{p^2}{2m} + V(x) = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x)$$

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} + V(x) \psi(x) = E \psi(x)$$

$$\psi_I(x) = A e^{kx} + B e^{-kx} ; \quad k = \sqrt{\frac{-2mE}{\hbar^2}} \quad (\psi_I(x) = A' \sinh kx + B' \cosh kx)$$

$$\psi_{II}(x) = C e^{ikx} + D e^{-ikx} ; \quad k = \sqrt{\frac{2m(E+V_0)}{\hbar^2}} \quad (\psi_{II}(x) = C' \sin kx + D' \cos kx)$$

$$\psi_{III}(x) = F e^{kx} + G e^{-kx}$$

$$\int_{-\infty}^{\infty} |\psi|^2 dx = 1 \Rightarrow B = F = 0$$

robni pogoji:

$$\psi_I(-\frac{a}{2}) = \psi_{II}(-\frac{a}{2}) \quad \psi_{II}(\frac{a}{2}) = \psi_{III}(\frac{a}{2})$$

$$\psi_I'(-\frac{a}{2}) = \psi_{II}'(-\frac{a}{2}) \quad \psi_{II}'(\frac{a}{2}) = \psi_{III}'(\frac{a}{2})$$

→ sistem enačb bo imel netrivialno rešitev $\Leftrightarrow$ determinanta koeficientov = 0

→ koef. so odv. od $E \rightarrow E_n, \psi_n$

→ zahtevno, najprej izrek:

predpostavimo: $V(x)$ sod

$$H\psi(x) = E\psi(x) \rightarrow x \rightarrow -x \rightarrow H\psi(-x) = E\psi(-x)$$

1) E nivo ni degeneriran

$$\psi(-x) = e^{i\alpha}\psi(x); \alpha \in \mathbb{R}$$

$$x \mapsto -x$$

$$\psi(x) = e^{i\alpha}\psi(-x) = e^{2i\alpha}\psi(x)$$

$$\rightarrow e^{2i\alpha} = 1 \rightarrow e^{i\alpha} = \pm 1$$

$\rightarrow \psi$ mora biti liha ali soda

2) E nivo je degeneriran

$$\psi(x) + \psi(-x) \text{ soda}$$

$$\psi(x) - \psi(-x) \text{ liha}$$

Nazaj na nalogo:

a) rešitev je soda:

$$\psi_I = Ae^{Kx}$$

$$\psi_{II} = B \cos kx$$

$$\psi_{III} = Ae^{-Kx}$$

robni pogoji:

$$Ae^{K\frac{a}{2}} = B \cos k \frac{a}{2}$$

$$-AKe^{K\frac{a}{2}} = -Bk \sin k \frac{a}{2}$$

$$K = k \tan k \frac{a}{2}$$

$\downarrow$

$$\tan \frac{Ka}{2} = \frac{Ka}{ka}$$

$$\tan \frac{u}{2} = \frac{\sqrt{u_0^2 - u^2}}{u} = \sqrt{\left(\frac{u_0}{u}\right)^2 - 1}$$

b) liha

$$\psi_I = -Ae^{Kx}$$

$$\psi_{II} = B \sin kx$$

$$\psi_{III} = Ae^{-Kx}$$

robni pogoji:

$$Ae^{-K\frac{a}{2}} = B \sin k \frac{a}{2}$$

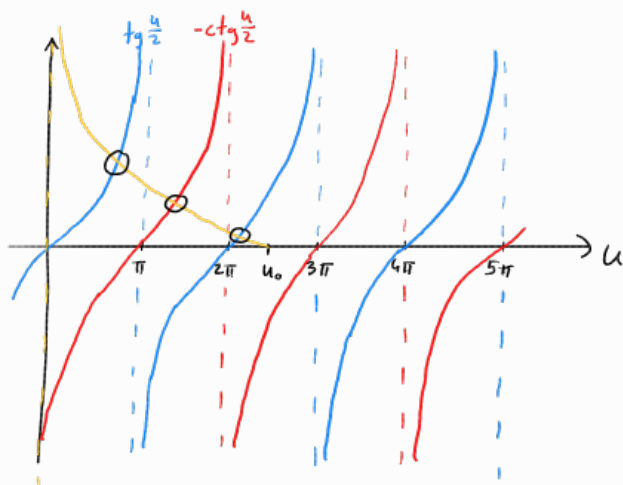
$$-AKe^{-K\frac{a}{2}} = Bk \cos k \frac{a}{2}$$

$$K = -k \cot k \frac{a}{2}$$

$\downarrow$

$$-\cot \frac{Ka}{2} = \frac{Ka}{ka}$$

$$-\cot \frac{u}{2} = \frac{\sqrt{u_0^2 - u^2}}{u} = \sqrt{\left(\frac{u_0}{u}\right)^2 - 1}$$



$\rightarrow$ za izbran u_0 imamo 3 vezana stanja

• 0. sn. je soda

• 1. vzb. je liha

• 2. vzb. je soda

št. vezanih stanj $\left[\frac{u_0}{\pi} \right] + 1$
 $\rightarrow$ celi del

Limitni primeri:

$$u_0 \rightarrow \infty$$

$$u = n\pi, n \in \mathbb{N}$$

$$\rightarrow L = \sqrt{\frac{2m(\epsilon + V_0)}{\hbar^2}}$$

$$\left(\frac{n\pi}{a}\right)^2 = \frac{2m(\epsilon + V_0)}{\hbar^2}$$

$$\rightarrow \epsilon = \frac{\hbar^2}{2m} \left(\frac{n\pi}{a}\right)^2 - V_0$$

$$V_0 \rightarrow \infty, a \rightarrow 0, V_0 a = \text{konst.}$$

$$u_0 \propto V_0 a^2 \rightarrow 0$$

$$\tan \frac{u}{2} = \sqrt{\left(\frac{u_0}{u}\right)^2 - 1} \quad u = u_0 - \xi, u_0 \rightarrow 0, \xi \ll u_0$$

$$\tan \frac{u_0 - \xi}{2} = \sqrt{\left(\frac{u_0}{u_0 - \xi}\right)^2 - 1} = \sqrt{\left(\frac{1}{1 - \xi/u_0}\right)^2 - 1}$$

$$\frac{u_0 - \xi}{2} = \sqrt{\left(1 + \frac{\xi}{u_0}\right)^2 - 1} = \sqrt{1 + 2\xi/u_0 - 1} = \sqrt{2\xi/u_0}$$

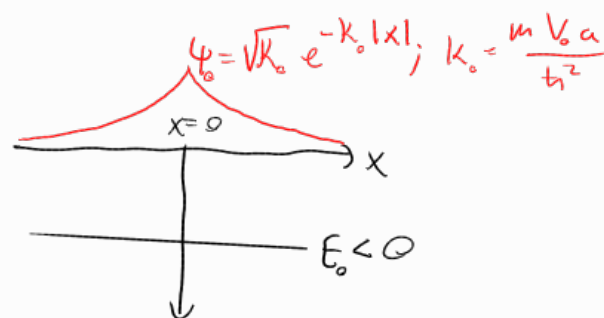
$$\xi = \left(\frac{u_0}{2}\right)^3 \rightarrow u = u_0 - \xi = u_0 - \frac{u_0^3}{8}$$

$$\rightarrow \left(\frac{u_0 - \frac{u_0^3}{8}}{a}\right)^2 = \frac{2m(\epsilon + V_0)}{\hbar^2} = \frac{1}{a^2} \left(u_0^2 - \frac{1}{4}u_0^4 + \left(\frac{u_0}{2}\right)^6\right)$$

$$\epsilon = \frac{\hbar^2}{2ma^2} \left(u_0^2 - \frac{1}{4}u_0^4\right) - V_0 \quad u_0^2 = \frac{2mV_0 a^2}{\hbar^2}$$

$$= \frac{\hbar^2}{2ma^2} \left(\frac{2ma^2 V_0}{\hbar^2} - \frac{1}{4} \frac{2m^2 V_0^2 a^4}{\hbar^4}\right) - V_0 =$$

$$= - \frac{m V_0^2 a^2}{2\hbar^2}$$



$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) - \lambda \delta(x) \psi(x) = E \psi(x) \quad \int_0^+ dx$$

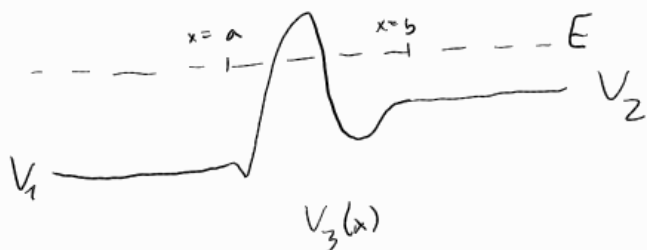
$$-\frac{\hbar^2}{2m} \psi'(x) \Big|_0^- - \lambda \psi(0) = E \int_0^+ \psi(x) dx = 0$$

$$\rightarrow \boxed{\psi'(0^+) - \psi'(0^-) = -\psi(0) \lambda \frac{2m}{\hbar^2} = -2K_0 \psi(0)} \rightarrow \text{če v potencialu nastopni } \delta \text{ funkcija, } \psi \text{ ne bo nikoli zv. odv.}$$

$\Rightarrow$ Robni pogoji za potencial z δ -funkcijo v a :

$$\psi'(a^+) - \psi'(a^-) = -2K_0 \psi(a)$$

$$\psi(a) = \psi(a^-) = \psi(a^+)$$



$$\psi_1 = A_1 e^{ik_1 x} + B_1 e^{-ik_1 x}$$

$$k_1 = \sqrt{\frac{2m(E-V_1)}{\hbar^2}}$$

$$\psi_2 = B_2 e^{ik_2 x} + A_2 e^{-ik_2 x}$$

$$k_2 = \sqrt{\frac{2m(E-V_2)}{\hbar^2}}$$

$$j(x) = \frac{\hbar}{2mi} [\psi^* \psi' - \psi (\psi^*)'] = \frac{\hbar}{4i} \ln [\psi^* \psi']$$

$$j_1(x) = \frac{\hbar}{4i} \ln [(A_1^* e^{-ik_1 x} + B_1^* e^{ik_1 x})(A_1 k_1 e^{ik_1 x} - B_1 k_1 e^{-ik_1 x})] =$$

$$= \frac{\hbar}{4i} \ln [A_1^2 k_1^2 - B_1^2 k_1^2 + A_1 B_1^* k_1 e^{2ik_1 x} - A_1^* B_1 k_1 e^{-2ik_1 x}] =$$

$$= \frac{\hbar k_1}{4i} (A_1^2 - B_1^2) = v_1 (A_1^2 - B_1^2)$$

$$j_2(x) = v_2 (B_2^2 - A_2^2)$$

$\rightarrow \psi_1, \psi_2$ normirano tako, da velja $v_1, v_2 = 1$

(v_1, v_2 pospravimo v konstante A, B)

$$j_1 = |A_1|^2 - |B_1|^2$$

$$j_2 = |B_2|^2 - |A_2|^2$$

Lahko zapisano $\psi_3(x) = C \psi(x) + D \chi(x)$

$$\left. \begin{array}{l} \psi_1(a) = \psi_3(a) \\ \psi_1'(a) = \psi_3'(a) \end{array} \right\} \rightarrow \text{dobimo } \underbrace{\begin{bmatrix} B_1 \\ B_2 \end{bmatrix}}_{\text{splošna matrika}(s)} = \underbrace{\begin{bmatrix} r & t' \\ t & r' \end{bmatrix}}_{\text{splošna matrika}(s)} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} r \\ t \end{bmatrix} = S \begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \begin{array}{l} j_1 = 1 - |r|^2 \\ j_2 = |t|^2 \end{array} \rightarrow \xrightarrow{\text{skalar}} \xrightarrow{\text{skalar}} |r|^2 + |t|^2 = 1$$

$$j_1 = j_2 \rightarrow |A_1|^2 + |A_2|^2 = |B_1|^2 + |B_2|^2$$

$$\text{ali } \begin{bmatrix} A_1^* & A_2^* \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = A^* A = B^* B = (SA)^* SA = A^* S^* S A$$

ker to velja $\forall A \Rightarrow S^* S = I \rightarrow S$ je unitarna

Če H realen: $A^* = S B^* \rightarrow B^* = S^* A^* \rightarrow B = S^T A \rightarrow S = S^T \rightarrow S$ je simetričen $\rightarrow S = \begin{bmatrix} r & t \\ t & r' \end{bmatrix}$

Poišči vse val. funkcije, za katere je $\delta x \delta p = \frac{\hbar}{2}$

(malička)

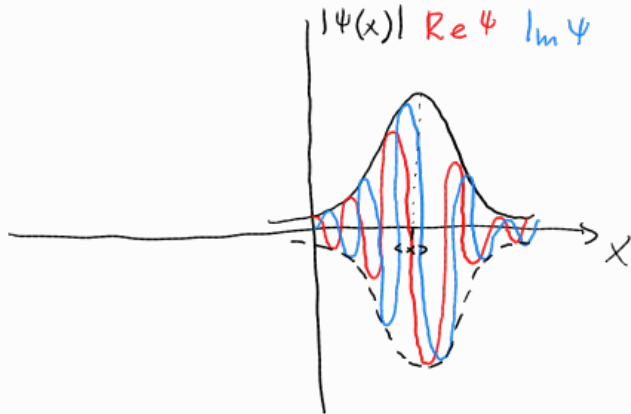
$$(p - \langle p \rangle) |\psi\rangle = i\mu (x - \langle x \rangle) |\psi\rangle, \mu \in \mathbb{R}$$

$$(-i\hbar \frac{d}{dx} - \langle p \rangle) \psi(x) = i\mu (x - \langle x \rangle) \psi(x)$$

$$\psi(x) = A e^{\frac{i}{\hbar} [\langle p \rangle - i\mu \langle x \rangle] x + i\mu \frac{x^2}{2}} = A e^{-\frac{\mu}{2\hbar} (x - \langle x \rangle)^2 + \frac{\langle x \rangle^2 \mu}{2\hbar} + \frac{i}{\hbar} \langle p \rangle x}$$

$$\int_{-\infty}^{\infty} dx |\psi(x)|^2 = 1 \Rightarrow \rho(x) = |\tilde{A}|^2 e^{-\frac{\mu}{\hbar} (x - \langle x \rangle)^2} = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x - \langle x \rangle)^2}{2\sigma^2}} \quad \left\{ \begin{array}{l} \text{pospravimo v } A \rightarrow \tilde{A} \\ \sigma > 0 \end{array} \right.$$

$$\rightarrow \psi(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x - \langle x \rangle)^2}{2\sigma^2} + \frac{i}{\hbar} \langle p \rangle x} \quad \left| \begin{array}{l} \text{Gaussov} \\ \text{val. paket} \end{array} \right.$$



$$\Psi(x,0) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\langle x \rangle)^2}{4\sigma^2}} e^{i\frac{\langle p \rangle}{\hbar}x}$$

$$H = \frac{p^2}{2m} + V(x), V(x) = 0$$

$$\langle x, t \rangle = \int \Psi^* x \Psi dx = ?$$

$$\langle p, t \rangle = ?$$

$$\delta x(t) = \sqrt{\langle x^2, t \rangle - \langle x, t \rangle^2} = ?$$

$$\delta p(t) = ?$$

$$\Psi_k = e^{ikx}, E_k = \frac{\hbar^2 k^2}{2m}, k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\rightarrow \Psi(x,0) = \int dk c(k) e^{ikx} \rightarrow \text{podobno kot Fourierova transf.}$$

$$\rightarrow c(k) = \frac{1}{2\pi} \int \Psi(x,0) e^{-ikx} dx$$

$$\rightarrow \Psi(x,t) = \int dk c(k) e^{-i\frac{E_k t}{\hbar}} e^{ikx} \rightarrow \text{zajetaj integral, iščemo boljše pot}$$

$$H \rightarrow \langle A, t \rangle = \langle \Psi, t | A | \Psi, t \rangle$$

$$|\Psi, t\rangle = e^{-i\frac{Ht}{\hbar}} |\Psi, 0\rangle \rightarrow \langle \Psi, t | = |\Psi, t\rangle^\dagger = \langle \Psi, 0 | e^{i\frac{Ht}{\hbar}}$$

operator čas.
razvoja

$$\langle A, t \rangle = \langle \Psi, 0 | \underbrace{e^{i\frac{Ht}{\hbar}} A e^{-i\frac{Ht}{\hbar}}}_{A(t)} | \Psi, 0 \rangle$$

očitno:

$$(\alpha A + \beta B)(t) = \alpha A(t) + \beta B(t)$$

$$(AB)(t) = A(t)B(t)$$

$$A^\dagger(t) = A(t)^\dagger$$

$$\frac{d}{dt} A(t) = e^{i\frac{Ht}{\hbar}} \frac{iH}{\hbar} A e^{-i\frac{Ht}{\hbar}} - e^{i\frac{Ht}{\hbar}} A e^{-i\frac{Ht}{\hbar}} \frac{iH}{\hbar}$$

$$= e^{i\frac{Ht}{\hbar}} \left(\frac{iH}{\hbar} A - A \frac{iH}{\hbar} \right) e^{-i\frac{Ht}{\hbar}} \quad \left| \begin{array}{l} \text{lahko zamenjamo, ker komutirata } ([f(B), B] = 0) \end{array} \right.$$

$$\frac{d}{dt} A(t) = \frac{i}{\hbar} e^{iHt/\hbar} [H, A] e^{-iHt/\hbar} = \frac{i}{\hbar} [H, A](t)$$

zác. pogoj: $A(0) = A$

x :

$$\frac{d}{dt} x(t) = \frac{i}{\hbar} [H, x](t) = \frac{i}{\hbar} \left[\frac{p^2}{2m}, x \right] = \frac{i}{\hbar 2m} [p^2, x] = \frac{i}{2\hbar m} (\underbrace{p[p, x] + [p, x]p}_{-i\hbar}) =$$

$$= \frac{i}{2\hbar m} [-2i\hbar p] = \frac{p(t)}{m}$$

zác. pogoj:

$$p(0) = p$$

$$x(0) = x$$

p :

$$\frac{d}{dt} p(t) = \frac{i}{\hbar} [H, p](t) = \frac{i}{2\hbar m} [p^2, p] = 0$$

$$\rightarrow p(t) = p$$

$$x(t) = \int \frac{p}{m} dt = \frac{p}{m} t + C \rightarrow C = x$$

$$\boxed{\langle x, t \rangle = \langle x(t), 0 \rangle = \langle \frac{p}{m} t + x, 0 \rangle = \frac{t}{m} \langle p, 0 \rangle + \langle x, 0 \rangle = \frac{t}{m} \langle p \rangle + \langle x \rangle}$$

$$\boxed{\langle p, t \rangle = \langle p(t), 0 \rangle = \langle p \rangle}$$

$$\delta p(t) = \sqrt{\langle p^2, t \rangle - \langle p, t \rangle^2} \quad (AB)(t) = A(t)B(t)$$

$$\langle p^2, t \rangle = \langle p^2(t), 0 \rangle = \langle p(t)^2, 0 \rangle = \langle p^2, 0 \rangle$$

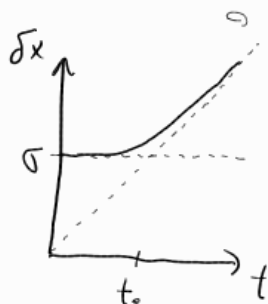
$$\begin{aligned} \langle A^\dagger \rangle &= \langle \psi | A^\dagger | \psi \rangle = \langle A \psi | \psi \rangle = \\ &= (\langle \psi | A | \psi \rangle)^* = A^* \\ \rightarrow \langle A + A^\dagger \rangle &= \langle A \rangle + \langle A \rangle^* = 2\text{Re}\langle A \rangle \end{aligned}$$

$$\boxed{\delta p(t) = \delta p(0) = \frac{\hbar}{2\sigma}}$$

$$\langle x^2, t \rangle = \dots = \langle x(t)^2, 0 \rangle = \langle \left(\frac{p}{m} t + x \right)^2, 0 \rangle = \langle \frac{p^2}{m^2} t^2 + \frac{t}{m} px + \frac{t}{m} xp + x^2, 0 \rangle$$

$$\begin{aligned} \rightarrow \delta^2 x &= \frac{t^2}{m^2} \left(\frac{\hbar}{2\sigma} \right)^2 + \frac{t}{m} \langle px + xp \rangle + \delta - 2 \frac{t}{m} \langle p, 0 \rangle \langle x, 0 \rangle = \\ &= \frac{t^2}{m^2} \left(\frac{\hbar}{2\sigma} \right)^2 + \sigma^2 + \underbrace{\frac{2t}{m} [\text{Re}\langle xp, 0 \rangle - \langle p, 0 \rangle \langle x, 0 \rangle]}_{=0} = \frac{t^2}{m^2} \left(\frac{\hbar}{2\sigma} \right)^2 + \sigma^2 \end{aligned}$$

$$\delta x(t) = \sigma \sqrt{1 + \frac{t^2}{m^2} \left(\frac{\hbar}{2\sigma^2} \right)^2}$$



$$\left(\frac{t_0 \hbar}{2m\sigma^2} \right)^2 = 1$$

$$t_0 = \frac{2m\sigma^2}{\hbar} = \frac{\hbar}{\frac{\hbar^2}{2m\sigma^2}} \rightarrow t_0 \in_0 = \hbar$$

